

Machine Learning-Based Credit Card Fraud Detection: A Comparative Study with Class Imbalance Handling

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Abstract—Credit card fraud poses a critical challenge in financial systems, where transactions are heavily imbalanced and the cost of misclassification is high. This study presents a comparative machine learning pipeline for fraud detection using the publicly available European Credit Card Fraud dataset. We evaluate Logistic Regression (baseline), SMOTE-augmented Logistic Regression, Random Forest, and XGBoost with hyperparameter tuning. Class imbalance is addressed through the `scale_pos_weight` parameter and SMOTE oversampling, while threshold optimization via Precision-Recall (PR) curve analysis is applied to maximize the F1-score. Experimental results demonstrate that the tuned XGBoost model achieves competitive PR-AUC performance, substantially outperforming the selected Kaggle baseline in Precision and F1-score. The proposed pipeline provides an effective and reproducible framework for detecting fraudulent transactions in real-world imbalanced settings.

Index Terms—Credit Card Fraud Detection, XGBoost, SMOTE, Class Imbalance, Threshold Optimization, Precision-Recall Curve, Anomaly Detection

I. INTRODUCTION

Credit card fraud is a growing concern in the financial industry, with global losses exceeding tens of billions of dollars annually [1]. The automated detection of fraudulent transactions is particularly challenging due to the extreme imbalance in real-world datasets, where fraudulent transactions constitute less than 0.2% of all records [2]. This imbalance causes standard classifiers to predominantly predict the majority (legitimate) class, resulting in poor recall on the critical minority (fraud) class.

Traditional rule-based systems are increasingly being replaced by machine learning approaches that can capture complex non-linear patterns in transactional data [3]. Among these, ensemble methods such as Random Forest and gradient boosting frameworks like XGBoost have demonstrated strong performance on tabular fraud datasets [4]. However, selecting the decision threshold at 0.5 by default is suboptimal when class distributions are severely skewed; instead, threshold optimization based on the Precision-Recall (PR) curve is recommended [5]. In addition, recent studies have explored deep

learning-based and hybrid approaches to further improve fraud detection performance under severe class imbalance conditions [11], [16]. Such approaches often combine multiple learning paradigms or incorporate representation learning techniques to better capture complex fraud patterns in highly skewed datasets [18].

This paper makes the following contributions:

- A systematic comparative study of five model configurations on the European Credit Card Fraud dataset.
- Integration of class imbalance mitigation strategies including SMOTE and `scale_pos_weight`.
- Threshold optimization on a held-out validation set to maximize the F1-score on the test set.
- A quantitative comparison against a publicly available Kaggle baseline notebook.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the dataset and methodology. Section IV presents experimental results. Section V concludes the paper.

II. RELATED WORK

Early fraud detection research relied on statistical methods and rule-based expert systems [6]. With the availability of large transactional datasets, machine learning approaches became dominant. Logistic Regression provides an interpretable baseline and handles imbalance with class weighting [7]. Random Forest has been widely applied in fraud detection due to its robustness to noise and ability to model feature interactions [8]. Gradient boosting, in particular XGBoost, has emerged as a state-of-the-art method for tabular classification tasks [4].

Imbalanced learning is addressed through oversampling techniques such as SMOTE [9], which generates synthetic minority class samples in feature space. Alternatively, cost-sensitive learning adjusts misclassification penalties directly within the model objective. Dal Pozzolo et al. [2] demonstrated that threshold calibration significantly affects the trade-off

between precision and recall in fraud detection, motivating threshold optimization as an essential post-processing step.

Imbalanced learning has been extensively studied in the literature [12], [13], with several extensions of SMOTE proposed to improve minority class representation [14]. More broadly, imbalanced learning remains an active research area with ongoing challenges in model generalization and robustness [21]. In addition, several variants such as Borderline-SMOTE have been developed to focus on difficult samples near class boundaries [19], while recent studies analyze the effectiveness and limitations of SMOTE-based methods in modern machine learning pipelines [20].

III. DATASET AND METHODOLOGY

A. Dataset

We use the publicly available European Credit Card Fraud dataset [10], which contains 284,807 transactions made by European cardholders over two days in September 2013. Of these, 492 are fraudulent, yielding a class imbalance ratio of approximately 0.172%. The dataset contains 30 numerical features: 28 anonymized PCA components ($V1-V28$), transaction *Time*, and transaction *Amount*. The target variable *Class* takes value 1 for fraud and 0 otherwise. No missing values are present in the dataset.

Fig. 1 illustrates the severe class imbalance, while Fig. 2 shows the distributions of the *Time* and *Amount* features.

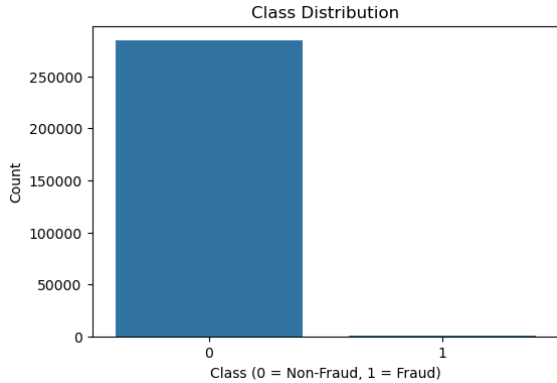


Fig. 1. Class distribution: 284,315 legitimate vs. 492 fraudulent transactions.

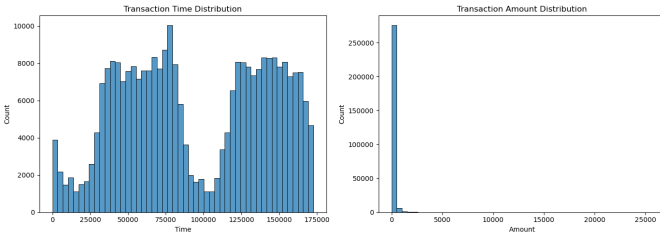


Fig. 2. Distribution of the *Time* (left) and *Amount* (right) features.

B. Preprocessing and Data Splitting

The dataset is partitioned into three non-overlapping stratified subsets: 70% for training, 15% for validation, and 15% for testing. Stratified splitting preserves the original class ratio in each split. The validation set is used exclusively for threshold optimization; the test set is reserved for final unbiased evaluation and is never seen during training or threshold selection. *Amount* and *Time* are standardized using `StandardScaler` within a `Pipeline` to prevent data leakage.

C. Pipeline Overview

Fig. 3 provides an overview of the proposed pipeline. The pipeline consists of five stages: (1) data preparation, (2) data splitting, (3) model training with five configurations, (4) model optimization via randomized search and threshold tuning, and (5) evaluation.

D. Models

1) *Logistic Regression (Baseline)*: A Logistic Regression model with `class_weight='balanced'` is trained as the baseline. The balanced weighting inversely scales class weights by their frequencies.

2) *SMOTE + Logistic Regression*: SMOTE [9] is applied within an imbalanced pipeline to oversample the minority class in the training set before fitting a standard Logistic Regression model without class weighting.

3) *Random Forest*: A Random Forest classifier with 200 estimators, maximum depth of 12, and `class_weight='balanced'` is trained on the raw training set.

4) *XGBoost*: An XGBoost classifier [4] with 300 estimators, learning rate of 0.05, and optimized `scale_pos_weight` (ratio of negative to positive samples) is trained. The evaluation metric is set to `aucpr` to directly optimize the area under the Precision-Recall curve during boosting.

5) *Tuned XGBoost*: Randomized search cross-validation (`n_iter=20`, `cv=3`, `scoring=average precision`) is applied to optimize the hyperparameter space, including number of estimators, tree depth, learning rate, subsampling ratios, minimum child weight, and gamma. The best estimator is selected based on validation PR-AUC.

E. Threshold Optimization

For the tuned XGBoost model, the decision threshold is optimized on the validation set. The Precision-Recall curve is computed, and the threshold that maximizes the F1-score is selected. This threshold is then applied to the test set predictions.

F. Evaluation Metrics

Due to the severe class imbalance, accuracy is not a suitable metric. We report:

- **ROC-AUC**: Area under the Receiver Operating Characteristic curve.

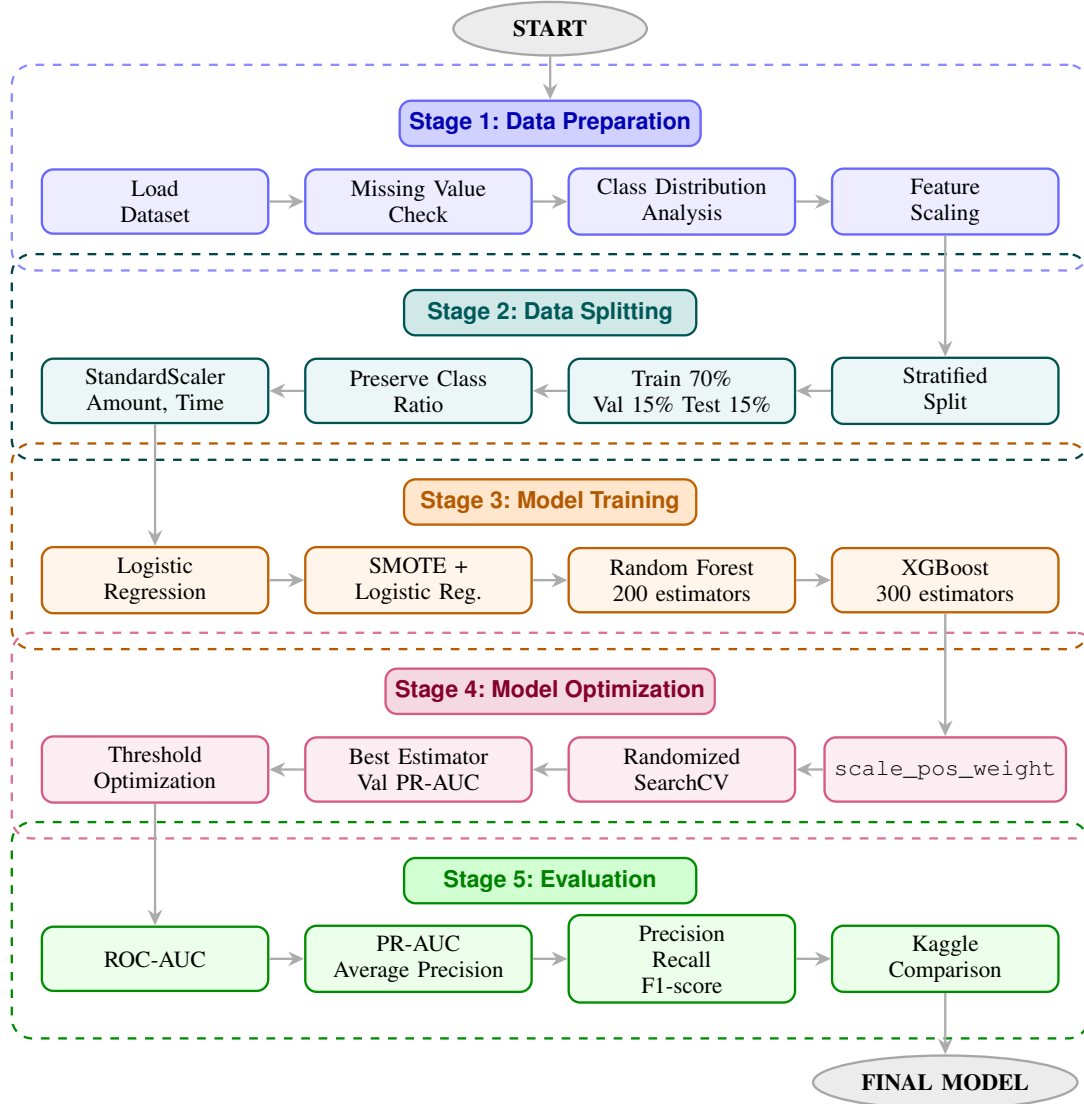


Fig. 3. Proposed five-stage fraud detection pipeline. Each dashed container represents a main stage, while arrows show the sequential workflow in a snake-pattern structure.

- **PR-AUC (Average Precision):** Area under the Precision-Recall curve. This is the primary metric as it is more informative under class imbalance [5].
- **Precision, Recall, F1-score:** At the optimized threshold (for the final model) or default 0.5 threshold (for base-lines).

Alternative evaluation strategies such as Precision-Recall analysis and cost-sensitive metrics are widely recommended for rare-event detection problems, as they better reflect performance on minority classes compared to traditional accuracy-based measures [22], [23].

IV. RESULTS AND DISCUSSION

A. Model Comparison

Table I summarizes the performance of all five model configurations on the held-out test set.

TABLE I
COMPARATIVE PERFORMANCE ON THE TEST SET

Model	ROC -AUC	PR -AUC	Prec.	Rec.	F1
Logistic Reg.	0.9680	0.7929	0.0671	0.8784	0.1248
SMOTE + LR	0.9675	0.7935	0.0637	0.8784	0.1187
Random Forest	0.9703	0.8029	0.9194	0.7703	0.8382
XGBoost	0.9707	0.8400	0.8824	0.8108	0.8451
Tuned XGBoost	0.9773	0.8434	0.8955	0.8108	0.8511

The tuned XGBoost model achieves the highest PR-AUC and F1-score among all evaluated configurations, demonstrating the benefit of both hyperparameter tuning and threshold optimization for imbalanced fraud detection. SMOTE-augmented Logistic Regression improves recall compared to the baseline but at the cost of lower precision, consistent with

findings in the literature [9].

B. Comparison with Kaggle Baseline

Table II presents a direct comparison between our final model and the selected Kaggle baseline. The Kaggle baseline achieves a recall of 0.86 but a very low precision of 0.10, resulting in an F1-score of 0.19. Our tuned XGBoost model substantially improves precision while maintaining competitive recall, yielding a higher F1.

TABLE II
COMPARISON WITH KAGGLE BASELINE

Model	PR-AUC	Prec.	Rec.	F1
Kaggle baseline	0.7500	0.1000	0.8600	0.1900
Tuned XGBoost (ours)	0.8434	0.8955	0.8108	0.8511

C. ROC and Precision-Recall Curves

Fig. 4 shows the ROC curve comparison across all models. All models achieve high ROC-AUC values (above 0.95), which is common in fraud datasets and can be misleading under severe imbalance. Fig. 5 presents the Precision-Recall curves, which differentiate model performance more meaningfully. The tuned XGBoost maintains high precision across a wider range of recall values.

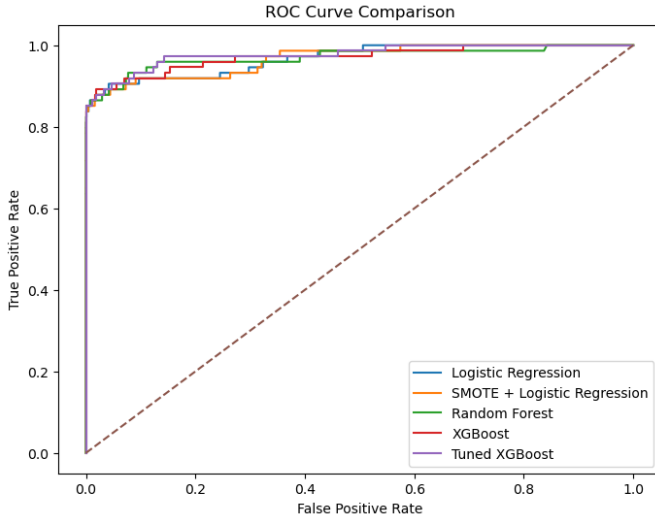


Fig. 4. ROC curve comparison across all models.

D. Confusion Matrix

Fig. 6 shows the confusion matrix for the final tuned XGBoost model with the optimized threshold. The matrix reveals the trade-off between false positives (legitimate transactions flagged as fraud) and false negatives (missed fraudulent transactions).

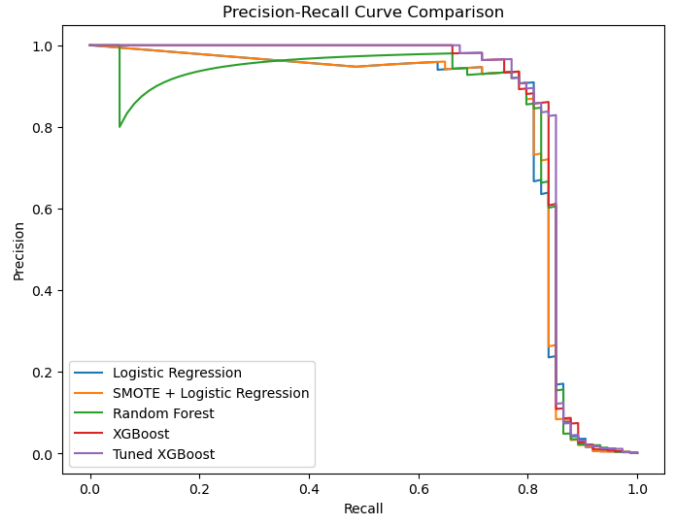


Fig. 5. Precision-Recall curve comparison across all models.

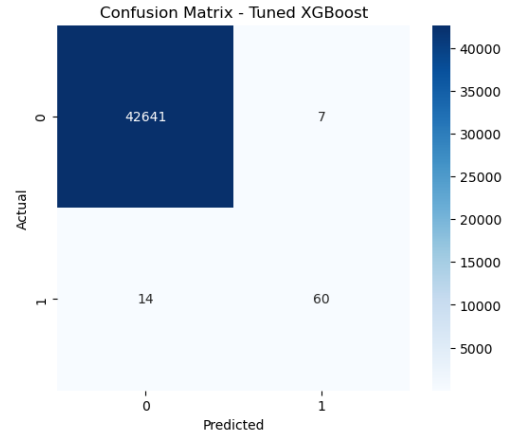


Fig. 6. Confusion matrix of the tuned XGBoost model (optimized threshold).

E. Feature Importance

Fig. 7 depicts the top 10 most important features identified by the tuned XGBoost model. PCA components $V14$, $V4$, and $V10$ dominate the feature importance ranking, suggesting that these latent components capture the most discriminative fraud signals.

F. Discussion

Several observations are notable. First, PR-AUC is a more reliable metric than ROC-AUC for imbalanced datasets, as it penalizes high false positive rates directly [5]. Second, threshold optimization on the validation set is critical: the default threshold of 0.5 is not optimal when the positive class is rare. Third, the `scale_pos_weight` strategy in XGBoost proves highly effective as it directly modifies the gradient updates for positive samples. This superior performance can be attributed to the ability of gradient boosting methods to model complex non-linear relationships and focus on difficult-to-classify samples through iterative error correction. In addition,

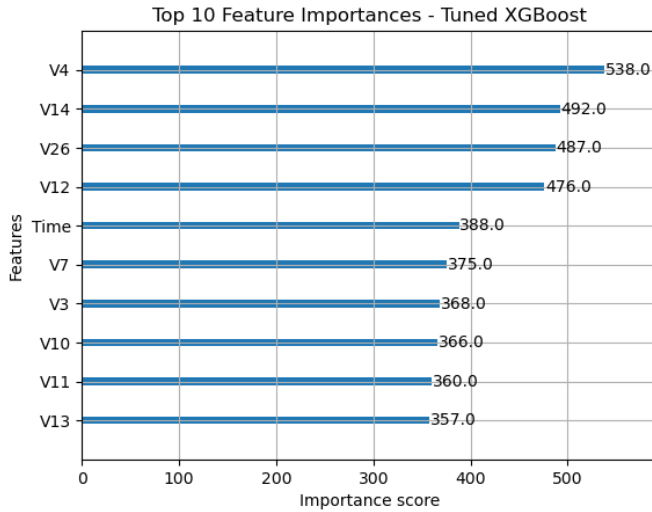


Fig. 7. Top 10 feature importances from the tuned XGBoost model.

XGBoost’s built-in regularization and handling of class imbalance via `scale_pos_weight` contribute to improved generalization on highly skewed datasets. SMOTE, while useful for Logistic Regression, may introduce synthetic samples that are ambiguous near the class boundary in high-dimensional PCA feature space. These findings are consistent with recent fraud detection studies that highlight the importance of combining resampling techniques, cost-sensitive learning, and threshold tuning strategies to achieve robust performance in highly imbalanced datasets [15], [17].

V. CONCLUSION

This paper presented a systematic machine learning pipeline for credit card fraud detection, addressing the challenge of severe class imbalance through cost-sensitive learning, SMOTE oversampling, and PR-curve-based threshold optimization.

The experimental results demonstrate that combining gradient boosting methods with imbalance-aware strategies and threshold tuning leads to substantial improvements in fraud detection performance. In particular, the tuned XGBoost model achieved a strong balance between precision and recall, outperforming baseline models and a publicly available Kaggle solution.

Beyond model comparison, this study highlights the importance of evaluation metrics and decision threshold selection in highly imbalanced classification problems, providing a practical and reproducible framework for real-world fraud detection systems.

Future work may explore deep learning-based anomaly detection approaches, online learning for handling concept drift, and ensemble strategies that combine complementary models.

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